LEMMA SES CHI TRIPLE

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter algebraization.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the document wrapper class, the xr-hyper wiring and the \external-document list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate was previously judged H2 by the initial-corpus stage; that judgement and its reason are reused verbatim and were not re-derived here. Original candidate key: S001-lemma-ses-chi-triple.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 0F23: sources/algebraization.tex, source lines 7996–7996. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 01YB: sources/coherent.tex, source lines 2621–2621. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00IN: sources/algebra.tex, source lines 12187–12187. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00IV: sources/algebra.tex, source lines 12394–12394. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Lemma 0.1. *We have* $\chi(\mathcal{G}, \mathcal{G}_0, \beta) = \chi(\mathcal{F}, \mathcal{F}_0, \alpha) + \chi(\mathcal{H}, \mathcal{H}_0, \gamma)$ *if*

$$0 \to (\mathcal{F}, \mathcal{F}_0, \alpha) \to (\mathcal{G}, \mathcal{G}_0, \beta) \to (\mathcal{H}, \mathcal{H}_0, \gamma) \to 0$$

is a short exact sequence of coherent triples.

Proof. Choose $\mathcal{G}', \beta', \beta'_0$ as in Lemma 0F23 for the triple $(\mathcal{G}, \mathcal{G}_0, \beta)$. Denote $j : U \to X$ the inclusion morphism. Let $\mathcal{F}' \subset \mathcal{G}'$ be the kernel of the composition

$$\mathcal{G}' \xrightarrow{\beta'} j_*\mathcal{G} \to j_*\mathcal{H}$$

Observe that $\mathcal{H}' = \mathcal{G}'/\mathcal{F}'$ is a coherent subsheaf of $j_*\mathcal{H}$ and hence $f : \mathcal{H}' \to \mathcal{H}'$ is injective. Hence by the snake lemma we obtain a short exact sequence

$$0 \to \mathcal{F}'/f\mathcal{F}' \to \mathcal{G}'/f\mathcal{G}' \to \mathcal{H}'/f\mathcal{H}' \to 0$$

We have isomorphisms $\alpha' : \mathcal{F}'|_U \to \mathcal{F}$, $\beta' : \mathcal{G}'|_U \to \mathcal{G}$, and $\gamma' : \mathcal{H}'|_U \to \mathcal{H}$ by construction. To finish the proof we'll need to construct maps $\alpha'_0 : \mathcal{F}'/f\mathcal{F}' \to \mathcal{F}_0$ and $\gamma'_0 : \mathcal{H}'/f\mathcal{H}' \to \mathcal{H}_0$ as in Lemma 0F23 and fitting into a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{F}'/f\mathcal{F}' & \longrightarrow & \mathcal{G}'/f\mathcal{G}' & \longrightarrow & \mathcal{H}'/f\mathcal{H}' & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \alpha'_0} & & \downarrow{\scriptstyle \beta'_0} & & \downarrow{\scriptstyle \gamma'_0} & & \\
0 & \longrightarrow & \mathcal{F}_0 & \longrightarrow & \mathcal{G}_0 & \longrightarrow & \mathcal{H}_0 & \longrightarrow & 0
\end{array}$$

However, this may not be possible with our initial choice of $\mathcal{G}'$. From the displayed diagram we see the obstruction is exactly the composition

$$\delta : \mathcal{F}'/f\mathcal{F}' \to \mathcal{G}'/f\mathcal{G}' \xrightarrow{\beta'_0} \mathcal{G}_0 \to \mathcal{H}_0$$

Note that the restriction of δ to U_0 is zero by our choice of $\mathcal{F}'$ and $\mathcal{H}'$. Hence by Cohomology of Schemes, Lemma 01YB there exists an $k > 0$ such that δ vanishes on $\mathfrak{m}^k \cdot (\mathcal{F}'/f\mathcal{F}')$. For $n > k$ set $\mathcal{G}'_n = \mathfrak{m}^n\mathcal{G}'$, $\mathcal{F}'_n = \mathcal{G}'_n \cap \mathcal{F}'$, and $\mathcal{H}'_n = \mathcal{G}'_n/\mathcal{F}'_n$. Observe that β'_0 can be composed with $\mathcal{G}'_n/f\mathcal{G}'_n \to \mathcal{G}'/f\mathcal{G}'$ to give a map $\beta'_{n,0} : \mathcal{G}'_n/f\mathcal{G}'_n \to \mathcal{G}_0$ as in Lemma 0F23. By Artin-Rees (Algebra, Lemma 00IN) we may choose n such that $\mathcal{F}'_n \subset \mathfrak{m}^k\mathcal{F}'$. As above the maps $f : \mathcal{F}'_n \to \mathcal{F}'_n$, $f : \mathcal{G}'_n \to \mathcal{G}'_n$, and $f : \mathcal{H}'_n \to \mathcal{H}'_n$ are injective and as above using the snake lemma we obtain a short exact sequence

$$0 \to \mathcal{F}'_n/f\mathcal{F}'_n \to \mathcal{G}'_n/f\mathcal{G}'_n \to \mathcal{H}'_n/f\mathcal{H}'_n \to 0$$

As above we have isomorphisms $\alpha'_n : \mathcal{F}'_n|_U \to \mathcal{F}$, $\beta'_n : \mathcal{G}'_n|_U \to \mathcal{G}$, and $\gamma'_n : \mathcal{H}'_n|_U \to \mathcal{H}$. We consider the obstruction

$$\delta_n : \mathcal{F}'_n/f\mathcal{F}'_n \to \mathcal{G}'_n/f\mathcal{G}'_n \xrightarrow{\beta'_{n,0}} \mathcal{G}_0 \to \mathcal{H}_0$$

as before. However, the commutative diagram

$$\begin{array}{ccccccc}
\mathcal{F}'_n/f\mathcal{F}'_n & \longrightarrow & \mathcal{G}'_n/f\mathcal{G}'_n & \xrightarrow[\beta'_{n,0}]{} & \mathcal{G}_0 & \longrightarrow & \mathcal{H}_0 \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
\mathcal{F}'/f\mathcal{F}' & \longrightarrow & \mathcal{G}'/f\mathcal{G}' & \xrightarrow{\beta'_0} & \mathcal{G}_0 & \longrightarrow & \mathcal{H}_0
\end{array}$$

our choice of n and our observation about δ show that $\delta_n = 0$. This produces the desired maps $\alpha'_{n,0} : \mathcal{F}'_n/f\mathcal{F}'_n \to \mathcal{F}_0$, and $\gamma'_{n,0} : \mathcal{H}'_n/f\mathcal{H}'_n \to \mathcal{H}_0$. OK, so we may use $\mathcal{F}'_n, \alpha'_n, \alpha'_{n,0}$, $\mathcal{G}'_n, \beta'_n, \beta'_{n,0}$, and $\mathcal{H}'_n, \gamma'_n, \gamma'_{n,0}$ to compute $\chi(\mathcal{F}, \mathcal{F}_0, \alpha)$, $\chi(\mathcal{G}, \mathcal{G}_0, \beta)$, and $\chi(\mathcal{H}, \mathcal{H}_0, \gamma)$. Now finally the lemma follows from an application of the snake lemma to

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{F}'_n/f\mathcal{F}'_n & \longrightarrow & \mathcal{G}'_n/f\mathcal{G}'_n & \longrightarrow & \mathcal{H}'_n/f\mathcal{H}'_n & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathcal{F}_0 & \longrightarrow & \mathcal{G}_0 & \longrightarrow & \mathcal{H}_0 & \longrightarrow & 0
\end{array}$$

and additivity of lengths (Algebra, Lemma 00IV). □